
SOIL PHYSICS

Quantification of Soil Macropores at Different Slope Positions under Alpine Meadow Using Computed Tomography in the Qinghai Lake Watershed, NE Qinghai–Tibet

Zong-Chao Li^{a, b}, Xia Hu^{a, c, *}, Xiao-Yan Li^{a, c}, Yong-Mei Huang^{a, c}, Xiu-Chen Wu^{a, c},
Pei Wang^{a, c}, and Lian-You Liu^{a, d}

^aKey Laboratory of Earth Surface Processes and Resource Ecology, Beijing Normal University, Beijing, 100875 China

^bChina National Environmental Monitoring Centre, Beijing, 100012 China

^cSchool of Natural Resources, Faculty of Geographical Science, Beijing Normal University, Beijing, 100875 China

^dAcademy of Disaster Reduction and Emergency Management, Beijing Normal University, Beijing, 100875 China

*e-mail: huxia@bnu.edu.cn

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Abstract—Soil macropores largely control fluid and solute transport and the runoff processes on the slopes. However, the characterization of soil macropores for different slope positions under alpine meadow conditions remains scarce, especially in the Qinghai Lake watershed. The objective of this study was to visualize and quantify the macropore characteristics of soils at different slope positions (i.e., the upper, middle and bottom slopes) in an alpine meadow in the Qinghai Lake watershed using X-ray computed tomography (CT). Nine soil cores (0–40 cm deep) were collected from the upper, middle and bottom slopes with three replicates per position, and cores were scanned with a GE HISPEED FX/I Medical Scanner. Macroporosity, length density, node density and tortuosity within the 40 cm soil profile were interpreted from the CT data to analyze the soil architecture. The results showed that the macroporosity decreased from the upper slope to the bottom slope. The macroporosity of soil on the upper slope was 50 times greater than that on the bottom slope, which can be attributed to the gravel in the former soil. The large number of continuous macropores and cracks concentrated in the deep soil layers (200–350 mm) of the upper slope can be attributed to the gravel layer. Most of the soil macropores under the middle and bottom slopes were distributed in the surface soil (0–150 mm), which can be attributed to the “mattic epipedon.” The soil macropores in the bottom slope, due to the well-developed root system, were more continuous and tubular than those in soils of the upper and middle slopes. It is expected that water can move preferentially through soil macropores on the upper and middle slopes compared to the bottom slope. Furthermore, due to its relatively low macroporosity the soil of the bottom slope contributes more to surface flow than those of middle and upper slopes.

Keywords: soil macropore, X-ray CT, slope position, alpine meadow, water

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INTRODUCTION

Macropores (i.e., earthworm burrows, root channels, fissures and interaggregate voids) represent only a small fraction of the total porosity in soil, but play a central role in many physical processes, such as fluid flow and solute transport [11, 12, 33]. In recent years, computed tomography (CT) scanning has been widely applied to analyze macropore structures [1, 5, 14, 27]. The geometrical and morphological parameters of complex macropore systems, such as their connectivity, tortuosity, porosity, and pore surface area, are very useful for characterizing macropore structures [2, 3, 10, 25, 34]. The macroporosity and macropore density of soil are also important in the analysis of macropore structures [17, 22]. It is well known that the formation of macropores is affected by wetting and drying cycles,

compaction, biological activities, plant roots, and cracks [17, 21, 35]. Furthermore, the formation of macropores is related to the physico-chemical properties of soil, such as the bulk density, soil moisture content, particle size distribution, and organic matter content. In addition, many researchers have reported that bulk density, soil moisture content, particle size distribution and organic matter content are related to slope position [10, 43, 44]. Hydrologically active macropores (pores larger than 1 mm diameter) represent a very small fraction of the total soil porosity but are the preferred flow path through forested hillslopes [31]. Zhang et al. indicated that total porosity is higher under lower slope than under upper slope [45]. Therefore, we hypothesize that the characteristics of soil macropores can be significantly different between slope positions.

Over the last three decades, X-ray computed tomography (CT) has gained increasing importance in soil science, especially in soil macropore sciences [8]. The X-ray CT technique has proven to be a very powerful and nondestructive technique for visualizing and quantifying soil macropores [8, 37]. Macroporosity and the distribution of macropores and coarse mesopores have also been interpreted using medical computed tomography [9, 32]. Combined with morphological mathematical tools, X-ray computed tomography has been applied to visualize and quantify the three-dimensional structures of soil macropores [6]. The network topology of soil macropores has been reconstructed in intact soil columns [13], and the number of macropore networks, as well as their area density, branch length, node density tortuosity, hydraulic radius, numerical density, and connectivity, have been calculated [1, 18, 33]. Zhang et al. and Hu et al. quantified the soil properties and macropore characteristics in uncultivated sandy land in Northwest China using helical medical X-ray computed tomography [39, 44]. These studies suggest that helical medical CT is a suitable method for quantifying soil macropores.

“Alpine meadow is the main ecosystem in the Qinghai Lake Watershed, and it accounts for 55.9% of the total land area” [36]. Slope land is one of the largest landforms in the Qinghai Lake Basin [28]. Preferential flow through macropores can affect the runoff from hillslopes [12, 16]. Furthermore, abundant rainfall and overland flow can cause variation in the soil structure of slopes [25]. Smith with coauthors provided models that included infiltration and water flow information for hillslope soil [29, 30]. However, studies of the interactions between soil macropores and slope positions are rare, particularly in the Qinghai-Tibet Plateau. Therefore, the objective of this study was to quantify the soil pore features of soils beneath different slope positions using CT to provide a better understanding of significant variation in soil macropore characteristics among different slope positions.

MATERIALS AND METHODS

Study Sites and Core Samples

This study (2014–2015) was conducted in a valley of the Shaliu River in the Qinghai Lake watershed, which is located in the northeastern region of the Qinghai-Tibet Plateau (37.575° N, 100.035° E, Fig. 2). This area is located in a semiarid cold high-altitude climate zone. The mean annual temperature and precipitation between 1957 and 2015 were 0.1°C and 400 mm, respectively, in the watershed. Approximately 70–80% of the annual precipitation occurs in summer and autumn. The winter is clear, dry and cold, whereas the summer is warm and wet [36]. The soil type of the study area is alpine meadow soil covered by alpine meadow vegetation, which is the main ecosystem in

the region, accounts for 55.9% of the Qinghai Lake watershed [42].

Nine intact soil columns (with three replicates) were sampled from the upper slope (I), the middle slope (II) and the bottom slope (III) in July 2014 (Fig. 2). The upper (I), middle (II) and bottom (III) slopes were 12°, 8°, and 16°, respectively. The distance between the upper slope and the middle slope was 200 m (altitude difference 50 m), and the distance between the middle slope and the bottom slope was 150 m (altitude difference 20 m). Three replicates were collected along a horizontal line at the same altitude in 5-m intervals. Soil cores were collected using a polyvinyl chloride (PVC) pipe, which was 100 mm in diameter and approximately 400 mm in length; a wedge-shaped iron pipe head (with a sharpened downward edge) was fixed to the bottom of the pipe. The aboveground shrub or grass was carefully clipped to ground level with minimum disturbance, and litter was completely removed to expose bare soil prior to soil sampling. The core was extracted by slowly and vertically pushing the cylinder into the soil; the penetration of the cylinder progressed centimeter by centimeter, and the soil around the bottom of the tube was carefully removed at each step to minimize mechanical stress during extraction. Meanwhile, the gravel around the bottom of the tube was carefully removed, and the pressure was increased in the gravel layer. After sampling, both ends of the PVC pipe were packed with wheat straw, and then blocked with matching PVC heads. Nine soil samples were wrapped in sponges and transported to the laboratory.

At each site, the soil profiles were divided into 3 soil horizons (0–100 mm, 100–200 mm and 200–400 mm), and soil samples were collected from each horizon for soil property analysis. The soil particle-size distribution was measured using a MasterSizer 2000. Soil particles were classified by size as sand (63 to 2000 µm), silt (2 to 63 µm) or clay (<2 µm) based on the World reference base for soil resources 2015. The organic carbon contents of these samples were determined using the potassium dichromate heating oxidation-volumetric method. The soil characteristics of the three soil horizons under different slope positions are listed in Table 1. The saturated hydraulic conductivity (K_{sat}) is amount of water in meters per unit time entering the soil, regardless of the types or values of forces or gradients. K_{sat} was measured at tensions of 10, 30, and 60 mm with a mini disk infiltrometer (Fig. 1).

CT Scanning and Image Analysis

A GE HISPEED FX/I Medical Scanner (Beijing Shijitan Hospital, China), which is a 64-slice spiral CT, was used to scan soil cores at an energy level of 140 kV and 300 mA with a scanning interval of 0.625 mm. The scanning time for each sample was 30 s. The scanning produced images (8-bit) with a slice thick-

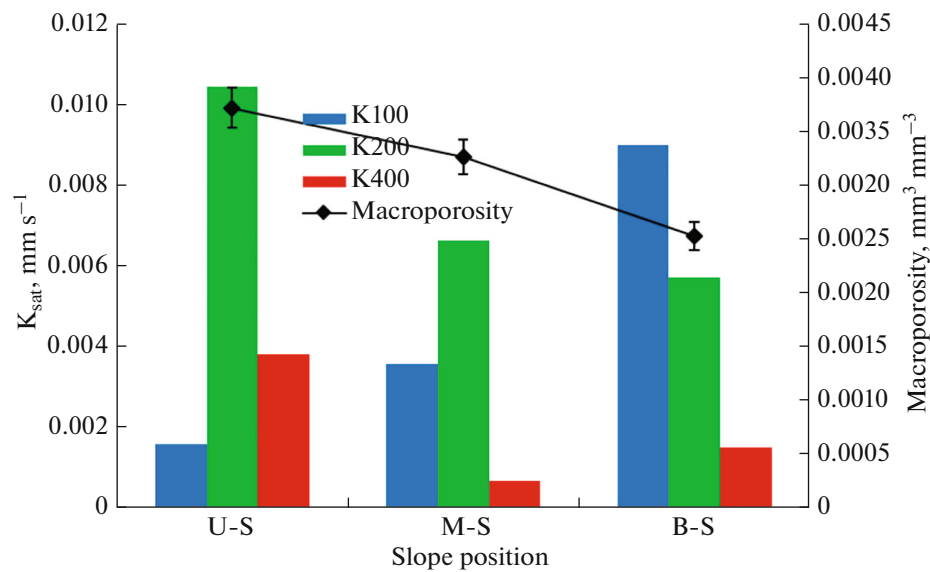


Fig. 1. Saturated hydraulic conductivity (K_{sat}) of the 0–100 mm soil layer (K_{100}), 100–200 mm soil layer (K_{200}) and 200–400 mm soil layer (K_{400}), as well as macroporosity of the topsoil (0–100 mm) in the U-S (upper slope), M-S (middle slope), and B-S (bottom slope).

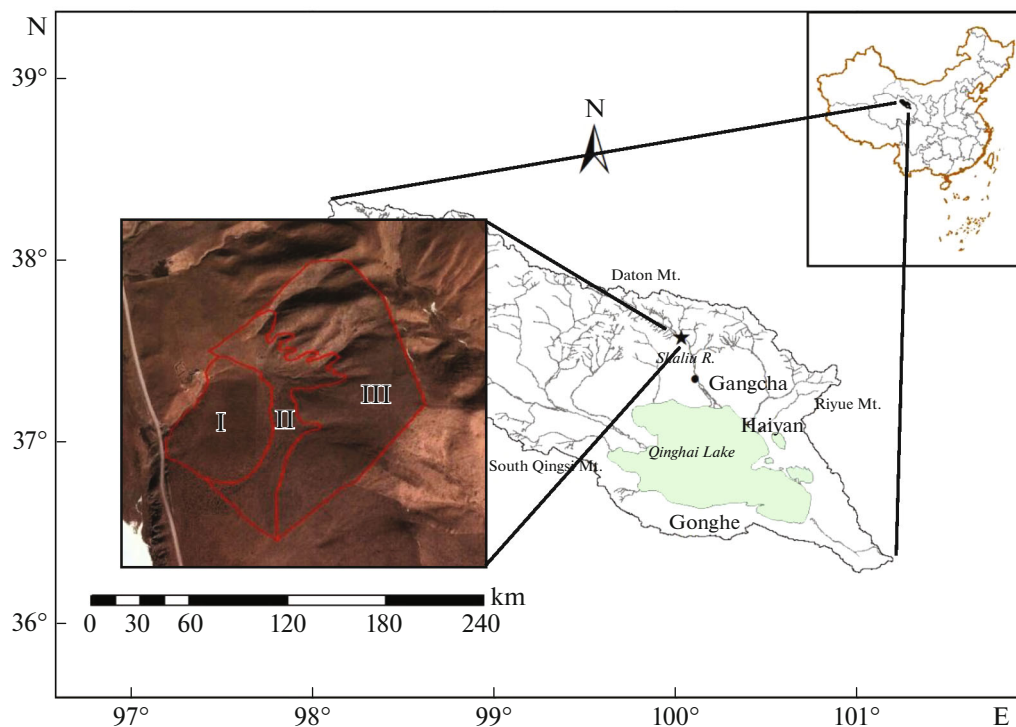


Fig. 2. Location of experimental site in the Qinghai Lake watershed. The star indicates the sampling site. Areas I, II, and III represent the bottom slope, middle slope and upper slope, respectively.

ness of 0.625 mm and 512×512 pixels per slice; thus, the resolution of voxel was $0.33 \times 0.33 \times 0.625$ mm in each reconstructed image. After scanning and reorientation, 640 slices were produced for each soil core, with an SNR (signal-to-noise ratio) of 2.76.

A Plexiglas cylinder with a known inner diameter (20 mm) was considered an artificial macropore. We inserted this cylinder into a soil core filled with soils that had been sieved to 2 mm and saturated with deionized water. The core was then scanned under the

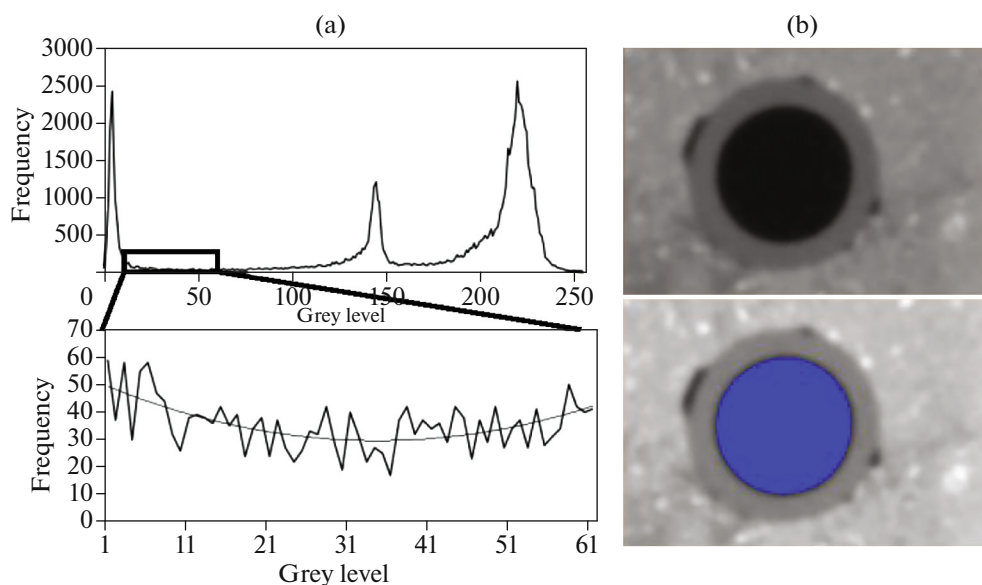


Fig. 3. Gray-value histogram produced using the thresholding method (a); grayscale image with segmented pores shown in blue (b).

previously described scanning conditions. First, the CT images were calibrated to Hounsfield units. The threshold value for the binarization of images was determined using an artificial macropore. A temporary threshold value of approximately 30–70 [33], which was the minimum point of the parabola, was obtained from the gray frequency histogram of the image (Fig. 3a). We calculated the macropore area using the Analyze Particles tool in Image J, version 1.50e (Fig. 3b). The threshold value was determined by comparing the calculated porosity with the measured porosity on an iterative loop until the difference was less than 1% [41]. The ultimate threshold value was 34; values lower than this threshold were identified as air-filled pores, and values greater than this threshold were identified as non-pores [7, 38]. The artificial macropore, as interpreted using the ultimate threshold value, is shown in the blue zone in Fig. 3b.

The images were processed and analyzed using AVIZO Fire® 9.0. A Gaussian filter was used to effectively eliminate the image noise and to improve the SNR. A cylindrical cropping tool was applied to obtain a region of interest (ROI) [4] to remove obvious disturbances caused by the sampling. The global threshold method was used in the image to achieve binarization processing. After undergoing binarization and segmentation, all macropores <1 mm in equivalent cylindrical diameter (ECD) were removed. Thus, only macropores with an ECD ≥ 1 mm were included in this study. To quantify the inter-connectivity of the macropores, a node was defined as an intersection where two branches of pores were connected [18]. Macropore length was the actual length of the longest path and other branches of a macropore network. Macropore vertical length was the vertical distance

Table 1. Soil properties of sample sites

Slope position	Altitude, m	Soil layer, mm	Organic content, g kg ⁻¹	Total phosphorus, g kg ⁻¹	Sand, %	Silt, %	Clay, %	Bulk density, g cm ⁻³
Upper slope	3633	0–100	115.00	0.77	41.12	55.00	3.88	0.84
		100–200	86.30	0.68	43.12	53.00	3.88	1.06
		200–400	64.10	0.64	40.12	56.00	5.88	1.09
Middle slope	3582	0–100	215.00	0.95	48.30	42.00	8.70	0.58
		100–200	213.00	0.95	47.30	43.00	9.80	0.86
		200–400	195.00	0.71	34.30	56.00	9.30	1.66
Bottom slope	3563	0–100	172.00	0.76	46.30	44.00	8.80	0.46
		100–200	201.00	0.67	38.30	52.00	9.90	0.44
		200–400	134.00	0.44	41.30	54.00	9.70	0.50

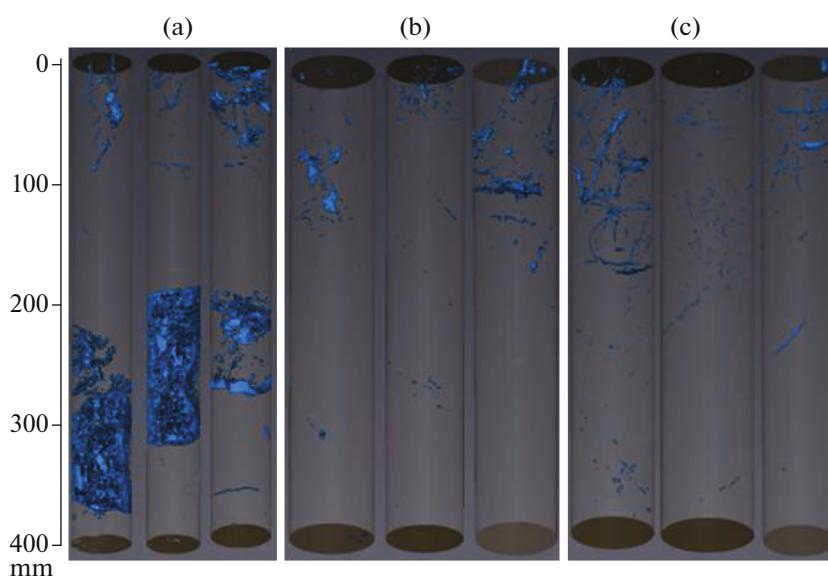


Fig. 4. Three-dimensional visualization of soil macropore networks in the soil columns of the (a) upper slope, (b) middle slope, and (c) bottom slope.

(straight-line) from the starting point to the ending point of a macropore network. The inclination of each macropore was characterized by the angle of the macropore from a vertical line. The macroporosity (volume of all macropores/volume of the ROI) and the number, volume, length, surface area, angle, branch (number and length), node number, and coordination number data for all macropores were calculated using Avizo 9.0. The surface area density, length density, number density, and node density were the macropore surface area, macropore length, and number of macropores and nodes in a unit volume respectively. The mean length, mean size, mean coordination number, and mean angle were the means of the length, volume, coordination number, and angle values of all macropores. Assuming that all macropores were cylindrical, the mean equivalent hydraulic radius of a macropore was calculated by the total volume and total length of the macropore. Furthermore, the mean tortuosity was calculated as the ratio of the total actual macropore length to the total straight-line distance of all the macropores in a certain volume [19].

Statistical Analyses

All statistical analyses were performed using SPSS software (version 13.0, SPSS Inc., Chicago, IL, USA). Differences in the measured parameters between sites of different slope positions were analyzed using two-way analysis of variance (ANOVA). Pearson's correlation coefficient (r) was used to test the correlations between the different macropore characteristics, as well as the correlations between the soil properties and macropore characteristics.

RESULTS

Visualization of Macropore Networks

Three-dimensional visualizations of the macropores in the nine soil columns are shown in Fig. 4. The spatial macropore characteristics clearly differ among soils at different slope positions. In the soil layer from 0 to 150 mm, the soil macropores under the upper slope were greater, deeper, and longer than those under the middle and bottom slopes. In addition, a large number of continuous macropores and cracks were concentrated in the deep soil layers (200–350 mm) of the upper slope due to the gravel layer. Most of the soil macropores under the middle and bottom slopes were distributed in the surface soil (0–150 mm). The macropores of soils under the bottom slope were more highly continuous and tubular than those of the soils under the upper and middle slopes.

Macroporosity and Macropore Number Density

The soil macropore characteristics of the nine columns are listed in Table 2. There were significant differences in the macropore characteristics among the three slope positions under the alpine meadow. The mean soil macroporosity decreased from the upper slope to the bottom slope. The mean macroporosity values of the upper slope, middle slope and bottom slope were $0.0501 \text{ m}^3 \text{ m}^{-3}$, $0.0031 \text{ m}^3 \text{ m}^{-3}$ and $0.0010 \text{ m}^3 \text{ m}^{-3}$, respectively.

Figure 5 shows the changes in macroporosity with depth in each soil core at the different slope positions. The curve is wavier for upper slope soil than for middle slope and bottom slope soil. In the soil of the upper slope, the macroporosity decreased from 0 to 10 mm,

Table 2. Macropore characteristics of all of the soil cores derived from CT images in the U-S (upper slope), M-S (middle slope), and B-S (bottom slope)

Column no.	Macro-porosity, $\text{m}^3 \text{m}^{-3}$	Macropore number density ($\times 10^5$), no. m^{-3}	Surface area density, $\text{m}^2 \text{m}^{-3}$	Length density, km m^{-3}	Node density ($\times 10^5$), no. m^{-3}	Mean macro-pore size, mm^3	Hydraulic radius, mm	Mean coordination number	Tortuosity	Mean angle, $^\circ$
U-S-1	0.0583	12.74	0.0964	21.54	38.95	45.70	0.93	3.06	1.55	54.60
U-S-2	0.0749	18.43	0.1456	40.06	75.35	40.57	0.77	3.34	1.47	49.53
U-S-3	0.0172	7.37	0.0278	4.51	8.81	23.32	1.10	2.46	1.23	60.29
Mean	0.0501 b (± 0.0171)	12.85 b (± 3.19)	0.0899 b (± 0.0348)	22.04 (± 10.27)	41.04 (± 19.24)	36.53 b (± 0.52)	0.93 (± 6.77)	2.95 b (± 0.1)	1.41 b (± 0.26)	54.81 (± 0.1)
M-S-1	0.0016	2.27	0.0044	0.72	1.64	6.95	0.84	1.81	1.20	49.45
M-S-2	0.0015	5.47	0.0049	0.53	2.18	2.77	0.95	1.14	1.18	54.09
M-S-3	0.0062	3.17	0.0144	2.40	5.80	19.38	0.91	2.05	1.26	65.81
Mean	0.0031 a (± 0.0016)	3.64 a (± 0.95)	0.0079 a (± 0.0033)	1.22 (± 0.59)	3.20 (± 1.31)	9.70 a (± 4.99)	0.90 (± 0.03)	1.67 a (± 0.27)	1.21 a (± 0.02)	56.45 (± 4.87)
B-S-1	0.0021	3.60	0.0059	1.00	2.34	5.87	0.82	1.54	1.16	54.57
B-S-2	0.0001	0.40	0.0003	0.03	0.11	1.84	0.87	1.00	1.09	59.60
B-S-3	0.0009	1.60	0.0023	0.31	0.64	5.61	0.96	1.17	1.08	53.71
Mean	0.0010 a (± 0.0006)	1.8671 a (± 0.9333)	0.0028 a (± 0.0033)	0.45 (± 0.29)	1.03 (± 0.67)	4.44 a (± 1.3)	0.88 (± 0.04)	1.24 a (± 0.16)	1.11 a (± 0.03)	55.96 (± 1.84)

Data group columns with the same letter are not significantly different at $p = 0.05$; the absence of letters indicates no significant differences in that property. Data in parentheses are standard deviations.

increased from 10 to 70 mm, and was relatively stable from 70 to 150 mm. Then the macroporosity fluctuated at 150–360 mm and reached its maximum value ($0.2 \text{ mm}^3 \text{ mm}^{-3}$). In contrast, for the middle slope and bottom slope soils, the macroporosity curves are relatively smooth, decreasing rapidly at soil depths of 0–50 mm before remaining stable at depths of 50–170 mm.

The mean macropore number density (i.e., the number of CT-derived macropore networks per unit volume) ranged from 1.87 to $12.85 (\times 10^5 \text{ m}^{-3})$. The soil of the upper slope had the largest macropore number density ($12.85 \times 10^5 \text{ m}^{-3}$) over the entire column, and was significantly greater than the macropore number densities of the middle and bottom slopes ($2.27 \times 10^5 \text{ m}^{-3}$ and $1.87 \times 10^5 \text{ m}^{-3}$, respectively).

Macropore Length Density, Macropore Vertical Length, Surface Area Density, Macropore Tortuosity and Inclination

The soil of the upper slope had the greatest macropore length density (22.04 km m^{-3}); in comparison, the middle slope and upper slope had a lower macropore length density (0.56 km m^{-3} and 2.30 km m^{-3} , respectively) (Table 2).

The macropore vertical lengths in the nine soil columns are shown in Fig. 7. Macropore connectivity

increased with the total vertical length of the macropores within same vertical length interval. Therefore, the total vertical length of macropores within same vertical length interval was ordered as follows: bottom slope > middle slope > upper slope. This indicated macropore connectivity in the vertical orientation decreased as follows: bottom slope > middle slope > upper slope. These results are consistent with the macropore network shown in Fig. 4.

The mean macropore surface area density of the soil of the upper slope was $6.53 \text{ mm}^2 \text{ mm}^{-3}$, which was approximately three times larger than the corresponding value of the bottom slope ($2.63 \text{ mm}^2 \text{ mm}^{-3}$) (Table 2). Similarly, the mean macropore size of the soil of the upper slope was 36.53 mm^3 , which was higher than that of the middle slope (9.70 mm^3) and bottom slope (4.44 mm^3). Furthermore, the mean hydraulic radius of the soil of the upper slope was 0.93 mm, which was also larger than that of the middle slope (0.90 mm) and bottom slope (0.88 mm).

The mean tortuosities were 1.26, 1.21 and 1.11 for soils under the upper slope, middle slope and bottom slope, respectively (Table 2). The tortuosity of soil macropores in the middle slope was approximately 1.11, which was lower than the tortuosity of soil macropores in the upper slope and middle slope; this occurred because there were fewer observed biopores

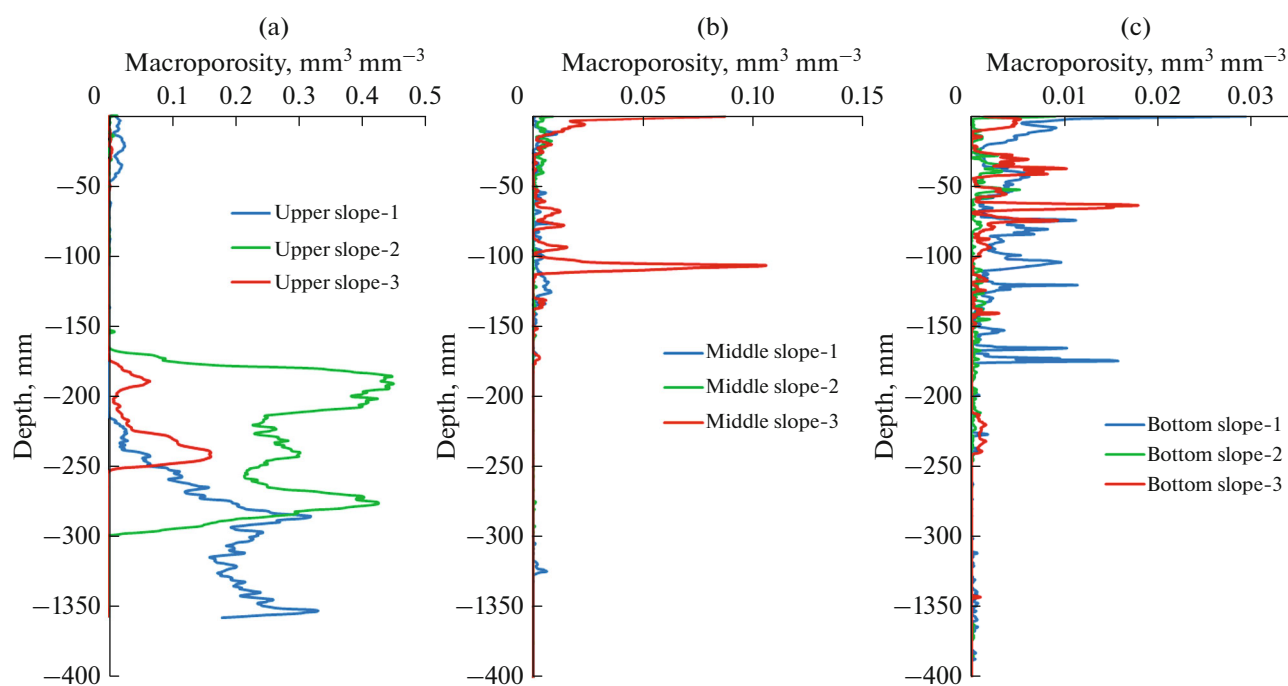


Fig. 5. Distribution of macroporosity along the depth of the soil column for the (a) upper slope, (b) middle slope, and (c) bottom slope. The numbers represent duplicate samples.

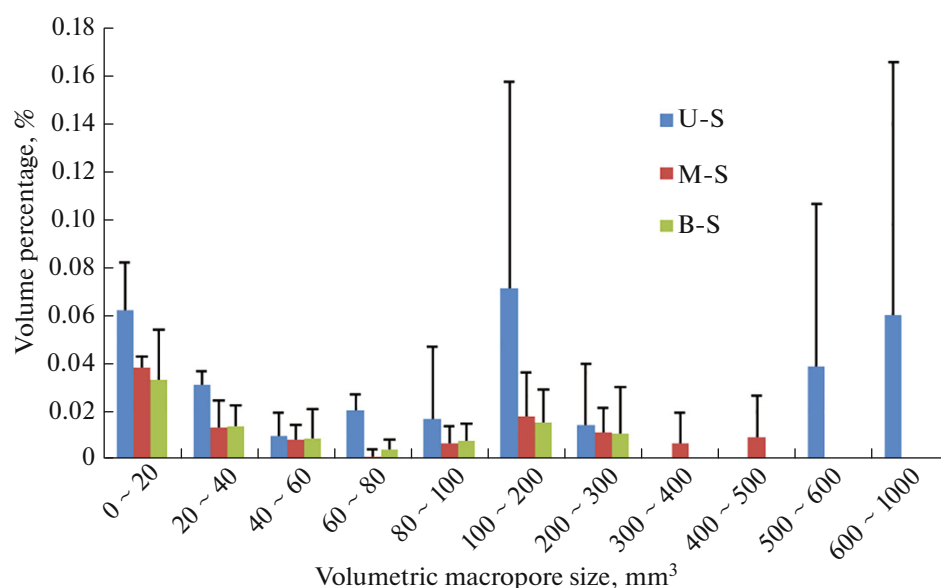


Fig. 6. Macropore size distribution (by volume) in the soil columns of the U-S (upper slope), M-S (middle slope), and B-S (bottom slope).

in the bottom slope than in the upper slope and middle slope.

Volume Macropore Size Distribution

Figure 6 shows the volumetric macropore size distribution (the proportion of different volumetric mac-

ropore volumes to the total macropore volume). The volume percentages of macropores within different volumetric macropore size ranges were calculated. The results indicated that more than 50% of the total macropore volume comprised macropores with volumes of <100 mm³ for soils of the middle slope (52.0%) and bottom slope (53.0%). More than 80% of

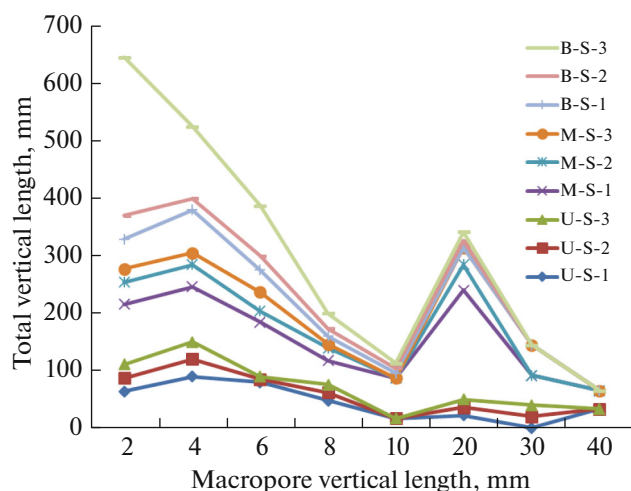


Fig. 7. Macropore vertical length distributions in the soil columns of the U-S (upper slope), M-S (middle slope), and B-S (bottom slope). Total vertical length is the sum of macropore vertical length Within the same vertical length interval (0–2, 2–4, 4–6, 6–8, 8–10, 10–20, 20–30 and 30–40 mm).

the total macropore volume comprised macropores with volumes of $>1000 \text{ mm}^3$ for the soil of the upper slope (83.0%). These results demonstrate that the upper slope induced greater macroporosity and more macropores than the middle slope and bottom slope; thus, there were more chances for macropores to form larger macropores at the upper slope than at the middle and bottom slopes.

DISCUSSION

The results showed that the macropores of soils under the bottom slope were more highly continuous

and tubular than those of the soils under the upper and middle slopes. Luo et al. indicated that soil macropores are highly continuous and cylindrical in shape are formed by roots [19]. Soil macropores formed by inter-aggregate macropores, such as those produced by freezing and thawing or by wetting and drying, are smaller, more random and less continuous than root-formed macropores [18]. Therefore, we expected that the relatively high macroporosity and highly tortuous paths in the soil under the middle slope might be due to the well-developed root system of this region. The macropores in the upper slope were formed by the gravel layer, which was less continuous than the root system.

The results are also consistent with the soil profiles. The results showed that soil macropores were mainly distributed in the 0 to 150 mm soil layer. Figure 8 shows the soil profiles of different slope positions (i.e., the upper, middle, and bottom slopes). The highest fine-root-biomass density was mainly in the 0–150 mm soil layer (Fig. 8), similar to the distribution of soil macropores (Fig. 4). We observed a large number of irregularly shaped macropores and cracks in the soil under the upper slope (Fig. 4a) and a large amount of gravel distributed at depths of 200–400 mm under the upper slope (Fig. 8a). According to Luo et al. [18], these small, randomly and discontinuously distributed macropores and cracks likely represent inter-aggregate macropores [18]. The mean macropore angle of the middle slope was the highest (56.45°), partly due to horizontally oriented *Potentilla fruticosa* shrub roots (Fig. 8b). Hence, the macropores of the soil under the middle slope were influenced by horizontal roots. The soil surface of alpine meadows contains a mixture of live and dead roots of different ages, which is called mattic epipedon [20]. Mattic epipedon was found in

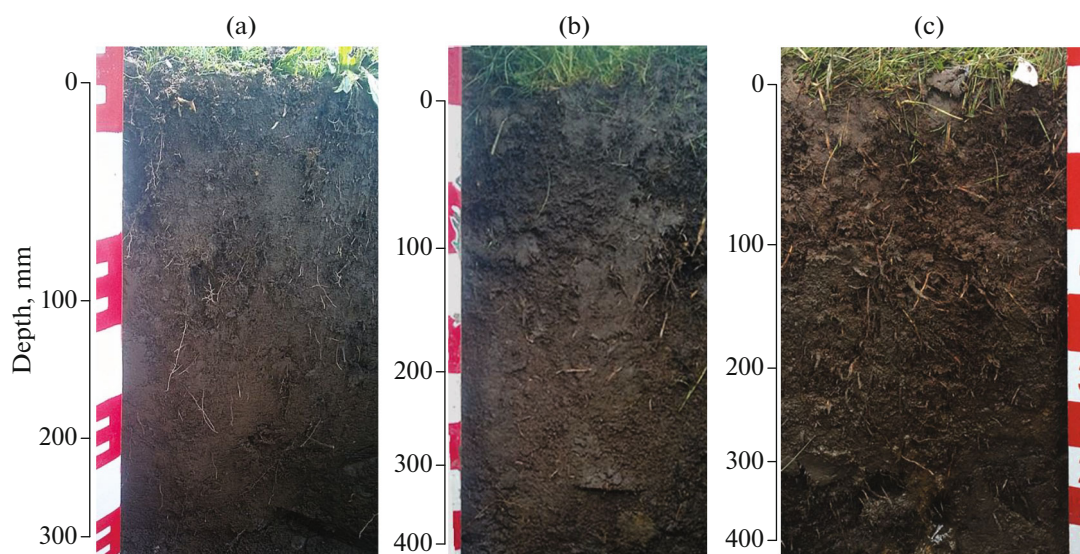


Fig. 8. Soil profiles of the (a) upper slope, (b) middle slope, and (c) bottom slope.

Table 3. Correlation (Pearson's correlation) matrix of macropore characteristics derived from the CT scans of soil samples

Properties	Macropore number density	Surface area density	Length density	Node density	Mean macropore size	Hydraulic radius	Mean coordination number	Tortuosity	Mean angle
	(A1)	(A2)	(A3)	(A4)	(A5)	(A6)	(A7)	(A8)	(A9)
Macroporosity	0.97**	0.99**	0.97**	0.97**	0.93**	−0.25	0.91**	0.92**	−0.34
A1		0.97**	0.96**	0.96**	0.88**	−0.19	0.89**	0.89**	−0.37
A2			0.99**	0.98**	0.91**	−0.31	0.90**	0.89**	−0.36
A3				0.99**	0.85**	−0.38	0.86**	0.85**	−0.40
A4					0.85**	−0.39	0.85**	0.84**	−0.40
A5						−0.03	0.96**	0.96**	−0.09
A6							−0.09	−0.18	0.47
A7								0.92**	−0.17
A8									−0.21

Probability values in parentheses are characterized by two significance levels (** < 0.01, * < 0.05).

the soil surface layer (0–100 mm) under the bottom slope (Fig. 8c). The mattic epipedon was deep and almost invisible at the bottom slope (Fig. 8). Although dense herb roots benefit macropore development, the macroporosity of the soil under an Alpine *Kobresia* meadow was less than that under *P. fruticosa* shrubs [38]. The soil macropores under the bottom slope were influenced by herb roots and freezing and thawing. Further analyses of the influence of roots and freezing and thawing on soil macropores will be conducted in the future. Clay was more beneficial to soil development, especially mattic epipedon development, than other types of soil particles. Table 1 shows that the clay particles of soil increased from the upper slope to the bottom slope. Mattic epipedon thickening from the upper slope to the bottom slope was essential for decreasing the soil macroporosity down the slope.

The connectivity of macropores affects their capacity to move water and thereby affects nutrient concentrations in runoff [16, 24, 40]. As expected, the hydraulic radius for soils under the upper slope and middle slope was larger, with mean values of 0.93 mm and 0.90 mm, respectively, than that for soils under the bottom slope (0.88 mm). We expect that water moves preferentially through macropores under the upper slope and middle slope, with less movement through soils under the bottom slope. The soil under the bottom slope would contribute more to surface flow than that under the upper slope and middle slope. This is consistent with the saturated hydraulic conductivity results. The Saturated hydraulic conductivity and macroporosity of the soil surface layer (0–100 mm) are shown in Fig. 1. The average soil macroporosity in the soil surface layer decreased from the upper slope middle slope to the bottom slope (0.0037, 0.0032 and 0.0025 mm³ mm^{−3}, respectively). In contrast, the saturated hydraulic conductivity of the soil surface layer increased from the upper slope to the bottom

slope (Fig. 1). Bulk density was mainly influenced by the soil matrix (clay content 8.80%) and grass roots. The soil bulk density of the bottom slope and the middle slope was smaller than that of the upper slope due to the mattic epipedon and intertwined grass roots in the soil surface layer. The saturated conductivity of the soil under the bottom slope tended to decrease with soil layer depth. Due to insufficient measurements of hydraulic conductivity, the reason for this pattern is not clear and needs further study [15, 26].

The macropore number density, soil macropores surface area density, mean macropore size, length density, mean coordination number and tortuosity were significantly correlated with macroporosity (Table 3). The high correlation of macroporosity with the node density ($r^2 = 0.97$), mean coordination number ($r^2 = 0.91$) and length density ($r^2 = 0.97$) suggests that the samples with higher values of CT-derived macroporosity also had more inter-connected macropores [33].

CONCLUSIONS

This study quantified the macropore characteristics (macroporosity, surface area density, length density, node density, mean coordination number, mean macropore size, hydraulic radius, mean tortuosity, and mean angle) of soil under different slope positions using X-ray CT in the Qinghai Lake Watershed of the NE Qinghai-Tibet Plateau. Our results demonstrated a clear distinction among the macropore characteristics of the three slope positions. The results showed that the macroporosity, macropore number density, surface area density, length density, node density, coordination number, tortuosity, and macropore size decreased significantly from the upper slope to the bottom slope. The results also showed that a large number of soil macropores were concentrated in the surface soil (0–150 mm) on the slope. Compared to

the soil of lower slopes, the soil of the upper slope had an increased number of soil macropores concentrated in the deep layer (200–350 mm) due to the gravel layer. The macropores under the middle slope were attributed to the large number of horizontal roots. In addition, the soil macropores of the soil under the bottom slope were mainly influenced by herb roots and by freezing and thawing. Roots and freeze–thaw processes significantly alter soil structural differences and influence flow and transport by generating macropore features in soils. It is expected that water preferentially moves through macropores under the upper and middle slopes, with less movement through bottom-slope macropores. The soil under the bottom slope contributes more to more surface flow than that under the middle and upper slopes.

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